

BSTBPC Backend: Change Report

Every endpoint under /api, and uploads stored in per-content folders

Item	Detail
Project	Bihar-Textbook-Backend (Node.js / Express 5 / Prisma / MySQL)
Date	24 September 2026
Scope	Backend only. Frontend changes are listed only where they are needed for the backend change to work.
Database schema	No change. No Prisma migration. Only stored path values change (via the optional script in section 5).
New dependencies	None. No npm install needed beyond what package.json already lists.

Contents

1. Summary
2. Why the change was needed
3. Part A: every endpoint under /api
4. Part B: uploads stored in per-content folders
5. Migration script for existing files
6. File-by-file change log
7. API contract changes for clients
8. Testing performed
9. Deployment checklist
10. Known issues and items not done

1. Summary

Two sets of changes were made to the backend:

A. Every endpoint now starts with /api

- All business routes were already mounted under /api. Three things were not, and have been moved: the landing page (/ to /api), Swagger (/api-docs to /api/docs), and uploaded files (/uploads/... is no longer served; files are served only from /api/uploads/...).
- Paths stored for new uploads now start with /api/uploads/ instead of /uploads/.
- Older rows that still hold /uploads/... are accepted everywhere the backend reads a path: the delete endpoint, the check for whether a file is still referenced, and file cleanup.
- Unknown /api/* routes now return a JSON 404 instead of Express's HTML page.
- Swagger works on the hosted server (a relative server entry was added) and the dashboard endpoints, which were missing because of a YAML error, now appear.

B. Uploads are stored in folders by content type

- Files used to go to uploads/images/, uploads/documents/ or uploads/videos/ depending only on file type. They now go to a folder for the content they belong to, e.g. uploads/notices/, uploads/circulars/, uploads/tenders/, uploads/books/covers/. There are 23 folders.
- The generic upload endpoints take a ?folder= query parameter, checked against a fixed allowlist. Unknown values are rejected with 400; no value means others/.
- Notice documents are moved automatically to notices/, circulars/ or tenders/ to match the notice's saved type and category, including when the type or category is changed later.
- A one-time script (npm run organize-uploads) moves existing files into the new folders and rewrites stored paths. It is a dry run unless --apply is passed.

2. Why the change was needed

On the hosted server (<https://bstbpc.bihar.gov.in>) only requests starting with /api/ are forwarded to this backend. Anything else goes to the website. That made uploaded files unreachable through the paths the backend was returning. Checked against production on 24 Sep 2026:

URL on https://bstbpc.bihar.gov.in	Result	Meaning
/uploads/images/1790248082188-...png	404 text/html	Not forwarded to the backend. This is the path the old frontend built, so leader photos were broken.
/api/uploads/images/1790248082188-...png	200 image/png	The same file, reachable because the path starts with /api.
/api/test	200 JSON	Backend reachable.
/api, /api/docs/	404	Production still runs the code from before this change; these start working after the backend is redeployed.

Keeping all uploads in three type-based folders also made it hard to tell which files belong to notices, tenders, circulars, books and so on, which is the reason for Part B.

3. Part A: every endpoint under /api

3.1 Endpoints that moved or were removed

Before	After	Notes
GET /	GET /api	Landing / status page (HTML with uptime and database status).
/api-docs	/api/docs	Swagger UI.
GET /uploads/*	Removed	Use /api/uploads/*.
GET /uploads/images/*, /uploads/image/*	Removed	Old aliases outside /api.
GET /uploads/documents/*, /uploads/document/*	Removed	Old aliases outside /api.
GET /uploads/videos/*, /uploads/video/*	Removed	Old aliases outside /api.
GET /api/uploads/images/*, /documents/*, /videos/*	Still served	No longer separate mounts; /api/uploads serves the whole uploads directory, so these paths still work.
GET /api/uploads/image/*, /document/*, /video/*	Kept	Singular aliases for the old images/, documents/, videos/ folders. Only needed until the migration script has run.
(none)	ANY /api/<unknown>	New JSON 404: { success: false, message: "Route not found: GET /api/..." }.

Anything outside /api (for example /) now returns Express's default 404.

3.2 Complete endpoint list (current)

Every route the backend serves. Paths and behaviour of these routes did not change except where noted in the last column.

Method	Path	Auth	Change
GET	/api	Public	Landing/status page. Moved from /.
GET	/api/docs	Public	Swagger UI. Moved from /api-docs.
GET	/api/test	Public	Health check.
GET	/api/uploads/<folder>/<file>	Public	Static uploaded files.
POST	/api/auth/signup	Dev key (x-dev-key)	
POST	/api/auth/reset-password	Dev key (x-dev-key)	
POST	/api/auth/login	Public	Rate limited.
POST	/api/auth/logout	Public	
GET	/api/auth/me	Admin (JWT)	
GET	/api/auth/profile	Admin (JWT)	
GET	/api/books	Public	
GET	/api/books/class/:classId	Public	
GET	/api/books/:id	Public	
POST	/api/books	Admin (JWT)	
PUT	/api/books/:id	Admin (JWT)	
DELETE	/api/books/:id	Admin (JWT)	
GET	/api/books/:id/chapters	Public	
POST	/api/books/:id/chapters	Admin (JWT)	

Method	Path	Auth	Change
PUT	/api/chapters/:id	Admin (JWT)	
DELETE	/api/chapters/:id	Admin (JWT)	
GET	/api/notices	Public	
GET	/api/notices/:id	Public	
POST	/api/notices	Admin (JWT)	Changed: document stored in notices/, circulars/ or tenders/ (section 4.4).
PUT	/api/notices/:id	Admin (JWT)	Changed: document moved when type/category changes (section 4.4).
DELETE	/api/notices/:id	Admin (JWT)	
GET	/api/sections	Public	
GET	/api/sections/:id	Public	
POST	/api/admin/sections/reorder	Admin (JWT)	
POST	/api/admin/sections	Admin (JWT)	Changed: uploaded files go to the module's folder (section 4.5).
PATCH	/api/admin/sections/:id	Admin (JWT)	Changed: uploaded files go to the module's folder (section 4.5).
DELETE	/api/admin/sections/:id	Admin (JWT)	
POST	/api/uploads/image	Admin (JWT)	Changed: new ?folder= parameter.
POST	/api/uploads/document	Admin (JWT)	Changed: new ?folder= parameter.
POST	/api/uploads/video	Admin (JWT)	Changed: new ?folder= parameter.
DELETE	/api/uploads	Admin (JWT)	Changed: accepts /api/uploads/... and legacy /uploads/... paths.
GET	/api/settings	Public	
GET	/api/settings/:key	Public	
PUT	/api/admin/settings/:key	Admin (JWT)	
POST	/api/admin/settings/:key/file	Admin (JWT)	Changed: file stored in csr-policy/ or printers/ by key.
DELETE	/api/admin/settings/:key/file	Admin (JWT)	Changed: removes files at both path forms.
GET	/api/getDirectory/:type	Public	
GET	/api/getDirectoryRow/:type/:id	Public	
POST	/api/admin/createDirectoryRow/:type	Admin (JWT)	
PATCH	/api/admin/updateDirectoryRow/:type/:id	Admin (JWT)	
DELETE	/api/admin/deleteDirectoryRow/:type/:id	Admin (JWT)	
GET	/api/admin/activities	Admin (JWT)	
POST	/api/admin/activities	Admin (JWT)	
PATCH	/api/admin/activities/:id/read	Admin (JWT)	
DELETE	/api/admin/activities/:id	Admin (JWT)	
DELETE	/api/admin/activities	Admin (JWT)	
GET	/api/admin/dashboard	Admin (JWT)	Now listed in Swagger.
GET	/api/admin/distribution	Admin (JWT)	Now listed in Swagger.
PUT	/api/admin/distribution/:year	Admin (JWT)	Now listed in Swagger.

3.3 Format of stored file paths

Every place the backend builds a path for a saved file now uses one helper, `toUploadUrl(file)`, which returns `/api/uploads/<folder>/<filename>`. Filenames are unchanged (`<timestamp>-<uuid>.<ext>`).

Where the path is produced	Before	After
POST <code>/api/uploads/{image document video}</code> (<code>data.path</code> , <code>data.url</code>)	<code>/uploads/images/x.png</code>	<code>/api/uploads/leaders/x.png</code>
POST/PUT <code>/api/notices</code> with a multipart document	<code>/uploads/documents/x.pdf</code>	<code>/api/uploads/circulars/x.pdf</code>
POST/PATCH <code>/api/admin/sections</code> with multipart files	<code>/uploads/images/x.jpg</code>	<code>/api/uploads/gallery/photos/x.jpg</code>
POST <code>/api/admin/settings/:key/file</code>	<code>/uploads/documents/x.pdf</code>	<code>/api/uploads/csr-policy/x.pdf</code>

3.4 Old `/uploads/...` paths already in the database

Existing rows keep working without a data migration:

- DELETE `/api/uploads` accepts a path starting with `/api/uploads/` or `/uploads/`. The error message is now `path must start with /api/uploads/`.
- Before deleting, the backend checks whether any row still references the file. That check now matches both forms of the path in `admin.avatarUrl`, `book.coverImageUrl`, `bookChapter.pdfUrl`, `notice.documentUrl`, `managingDirectorMessage.photoUrl`, `section.imageUrl/documentUrl/videoUrl` and `setting.settingValue`.
- Removing a replaced or deleted file (notices, sections, settings) resolves both forms, with the same guard against paths that escape the uploads directory.
- The backend no longer serves `/uploads/...`. The frontend's `fileUrl()` turns an old `/uploads/x` into `<API host>/api/uploads/x`, so old rows display correctly. The script in section 5 can rewrite them for good.

3.5 Swagger

- Served at `/api/docs` (was `/api-docs`).
- `servers` now lists `/` ("Current Server") first, then `http://localhost:3000`. "Try it out" works on the hosted server as well as locally.
- `src/routes/dashboardRoutes.js`: a summary containing `{ months: [...] }` was not quoted and broke the YAML, so the three dashboard endpoints were missing. It is quoted now. The spec went from 28 to 31 paths, all starting with `/api/`.
- The three upload endpoints document the new `folder` query parameter. Descriptions say files are stored in `uploads/{folder}/`. The delete example path is now `/api/uploads/notices/...`

3.6 Other small changes

- Landing page link to Swagger points to `/api/docs`.
- README: the two mentions of `/api-docs` now say `/api/docs`.
- Startup banner shows `http://localhost:<PORT>/api`.
- Comments that gave `/uploads/...` examples were updated (`common.js`, `uploads.js`, `controllers`).

4. Part B: uploads stored in per-content folders

4.1 Folder layout

Before uploads/ images/ documents/ videos/	(all images) (all PDFs/DOCs) (all MP4s)	After uploads/ notices/ books/covers/ leaders/ past-mds/ md-message/ gallery/photos/ tools-resources/ opmp/ others/	circulars/ books/chapters/ board-members/ employees/ csr-policy/ gallery/videos/ latest-initiatives/ registration-forms/	tenders/ officers/ admins/ printers/ rti/ gallery/press/
--	---	---	---	--

All 23 folders are created automatically when the server starts. The server no longer creates images/, documents/ or videos/; any that already exist are left alone until the migration script empties and removes them.

4.2 What goes in each folder

Folder	Contents	Source of the folder name
notices	Notice documents (type Notice, any category except Circular)	Frontend Notices page; backend moves it on save
circulars	Notice documents with category "Circular" (case-insensitive)	Same as above
tenders	Documents of notices with type Tender	Same as above
books/covers	Book cover images	Frontend Books page
books/chapters	Chapter PDFs	Frontend Books page
leaders	Leader photos (directory type leader)	Frontend Leaders page
board-members	Board member files (directory type board_member)	Website editor module ku-board
officers	Officer files (directory type officer)	Migration script only (no upload screen yet)
past-mds	Past MD files (directory type past_md)	Website editor module ku-list-md
employees	Employee files (directory type employee)	Migration script only
admins	Admin avatars	Migration script only
md-message	MD message photo	Website editor MD message page; setting key md_message
csr-policy	CSR policy PDF	CSR page; setting keys csr_policy_doc and csr_policy_content; section module csr
printers	Printer registry PDF	Setting key printer_registry_doc
rti	RTI files	Setting key dc-rti; website editor module dc-rti
gallery/photos	Photo gallery	Section module gl-photo
gallery/videos	Uploaded videos and their thumbnails	Section module gl-video
gallery/press	Press release images	Section module gl-press
tools-resources	Tools and Resources	Section module tr
latest-initiatives	Latest Initiatives	Section module cl
opmp	One Platform Many Possibilities	Section module opmp
registration-forms	Registration forms	Section module dc-reg-forms
others	Anything uploaded without a folder	Default when ?folder= is missing

4.3 How each endpoint picks the folder

Endpoint	How the folder is chosen
POST /api/uploads/image document video	?folder= query parameter. It must be one of the 23 names; anything else returns 400 with the full list. Missing means others.
POST /api/admin/settings/:key/file	From the key: csr_policy_doc to csr-policy, printer_registry_doc to printers. (Only these two keys are accepted by this route's validator.)
POST/PUT /api/notices (multipart document)	Saved to notices/ first, then moved to the folder for the saved type/category (4.4).
POST/PATCH /api/admin/sections (multipart image/document/video)	Saved to others/ first, then moved to the section module's folder (4.5).
POST/PUT /api/notices with documentUrl in JSON	The file (usually uploaded just before through /api/uploads/document) is moved to the right folder on save (4.4).

Mechanism: a new middleware `setUpUploadFolder(resolve)` runs before multer and sets `req.uploadFolder`. Multer's destination saves into `uploads/<req.uploadFolder>`, falling back to `others`. It replaces the old rule that picked `images/`, `documents/` or `videos/` from the file type.

4.4 Notice documents follow the notice's type and category

Rule (`noticeUploadFolder(type, category)`): type `Tender` goes to `tenders`; otherwise category `Circular` (case-insensitive, trimmed) goes to `circulars`; everything else goes to `notices`.

- **Create** (`createNotice`): the document (multipart upload or `documentUrl`) is moved to the matching folder before the row is written, and the row stores the new path.
- **Update** (`updateNotice`): the effective type, category and document are the new values if sent, otherwise the stored ones. If the document is not in the matching folder it is moved and the new path is saved. Example: changing a notice from `Circular` to `Tender` moves its PDF from `circulars/` to `tenders/` even if no new file was sent.
- **Rollback**: if the database write fails after a move, the file is renamed back to where it was (a warning is logged if that fails). For multipart uploads the existing cleanup then deletes the new file as before.
- **Replaced documents**: the old file is deleted only when a different document was sent, not when the same file was just moved.
- External links (not `/api/uploads` or `/uploads`) and files that no longer exist on disk are left unchanged.

4.5 Section files follow the section's module

`createSection` and `updateSection` call a new helper `moveReqFilesToModuleFolder(files, module)`. It moves every multipart image, document and video to `SECTION_UPLOAD_FOLDERS[module]` before building the stored paths. On update the module is the new one if sent, otherwise the stored one. Modules without a mapping keep their files in `others/`. Sections saved with JSON paths from `/api/uploads/*` (what the frontend does) already have the folder chosen at upload time.

4.6 Security and validation

- Folder names come only from a fixed allowlist, so a client cannot write outside the uploads directory. For example `?folder=../etc` returns 400.
- All path resolution still checks that the result stays inside `UPLOADS_DIRECTORY`; `/api/uploads/../../../../etc/passwd` resolves to null and is refused.
- File type and size limits are unchanged: images JPG/PNG/WEBP up to 2 MB, documents PDF/DOC/DOCX up to 20 MB, videos MP4 up to 50 MB, notice PDFs up to 10 MB, section uploads up to 50 MB overall.
- Upload endpoints still require an admin JWT.

4.7 Exports in src/middlewares/uploads.js

Export	Status	Purpose
UPLOAD_FOLDERS	New	The allowlist of 23 folder names.
DEFAULT_UPLOAD_FOLDER	New	"others".
SECTION_UPLOAD_FOLDERS	New	Section module to folder (opmp, tr, cl, csr, gl-photo, gl-video, gl-press, dc-reg-forms).
DIRECTORY_UPLOAD_FOLDERS	New	Directory type to folder (leader, board_member, officer, past_md, employee).
SETTING_UPLOAD_FOLDERS	New	Setting key to folder (csr_policy_doc, csr_policy_content, printer_registry_doc, md_message, dc-rti).
noticeUploadFolder(type, category)	New	Folder for a notice document (section 4.4).
setUpUploadFolder(resolve)	New	Middleware that sets req.uploadFolder; 400 for unknown names.
UPLOADS_URL_PREFIX	New	"/api/uploads/".
toUploadUrl(file)	New	Public path for a file multer saved.
uploadRelativePath(url)	New	Path inside uploads/ for a new or legacy URL, else null.
uploadUrlVariants(url)	New	Both stored forms of a URL, for reference checks.
getLocalUploadPath(url)	New	Safe absolute path, or null if outside uploads/.
moveUploadToFolder(url, folder)	New	Moves a stored file to a folder and returns the new URL.
moveUploadedFile(file, folder)	New	Moves a file multer just saved and updates file.path.
UPLOADS_IMAGES_DIR, UPLOADS_DOCUMENTS_DIR, UPLOADS_VIDEOS_DIR	Removed	Nothing else used them.
storage.destination	Changed	Uses req.uploadFolder instead of the file type.
UPLOADS_DIRECTORY, multer instances, handleUploadError	Unchanged	

5. Migration script for existing files

New file `scripts/organize-uploads.js`, run with the new npm script `organize-uploads`. Run it from the backend folder, where `uploads/` is.

```
npm run organize-uploads          # dry run: prints the plan, changes nothing
npm run organize-uploads -- --apply # moves files and updates the database
```

What it does

- Reads every column that stores an upload path and works out the folder for each file:

Table.column	Folder
<code>notice.documentUrl</code>	notices / circulars / tenders (by type and category)
<code>book.coverImageUrl</code>	books/covers
<code>bookChapter.pdfUrl</code>	books/chapters
<code>directory.imageUrl</code>	by directory type (leaders, board-members, officers, past-mds, employees; others if unknown)
<code>section.imageUrl</code> , <code>documentUrl</code> , <code>videoUrl</code>	by module (see 4.2; others if unknown)
<code>admin.avatarUrl</code>	admins
<code>managingDirectorMessage.imageUrl</code>	md-message
<code>setting.settingValue</code>	by key (see 4.7). Paths inside JSON values (<code>md_message.photo</code> , <code>csr_policy_content.pdfUrl</code>) are found and rewritten in place.

- Rewrites every local path to `/api/uploads/<folder>/<file>`, which also removes the old `/uploads/...` form from the database.
- If two rows point to the same file, the first one found decides the folder and both rows get the same new path.
- Referenced files that are missing on disk are listed and their rows are left unchanged.
- Files in `images/`, `documents/` or `videos/` that no row references are listed and left where they are, so nothing unused is deleted.
- With `--apply`: moves the files first, then updates all rows in one database transaction (60-second timeout, one update at a time). If anything fails, the files moved so far are moved back and the process exits with code 1.
- After a successful run, `images/`, `documents/` and `videos/` are deleted if they are empty (ignoring `.DS_Store`).

Dry run on the local development database (24 Sep 2026)

```
Files to move (1):
  images/1789832210596-5f55aab3-...jpg -> leaders/

Database rows to update (1):
  directory #68.imageUrl: /uploads/images/1789832210596-...jpg
                        -> /api/uploads/leaders/1789832210596-...jpg

Files in old folders that no row references, left in place (11):
  images/... (10 files)  documents/... (1 file)

Dry run - nothing changed. Re-run with --apply to move files and update the database.
```

`--apply` has not been run on either the local or the production database.

6. File-by-file change log

All paths are relative to Bihar-Textbook-Backend/.

server.js

- Landing page route moved from GET / to GET /api.
- Swagger mounted at /api/docs instead of /api-docs.
- Static file serving reduced to /api/uploads (whole uploads directory) plus the singular aliases /api/uploads/image, /api/uploads/document, /api/uploads/video. The 8 non-/api mounts (/uploads, /uploads/images, /uploads/image, /uploads/documents, /uploads/document, /uploads/videos, /uploads/video) and the separate plural /api mounts were removed.
- New JSON 404 handler for unknown /api routes, registered after all routers and before the upload error handler and global error handler.
- Startup banner shows http://localhost:PORT/api.

src/config/swagger.js

- Added servers entry { url: "/", description: "Current Server" } before the localhost entry.

src/middlewares/uploads.js

- Added the folder allowlist, folder maps, notice folder rule, setUploadFolder middleware and path helpers listed in 4.7.
- Creates all 23 folders at startup; no longer creates images/, documents/, videos/.
- Removed UPLOADS_IMAGES_DIR, UPLOADS_DOCUMENTS_DIR, UPLOADS_VIDEOS_DIR.
- Multer destination now uses req.uploadFolder (default others).
- Imports rename from node:fs/promises.

src/controllers/uploadController.js

- buildFileResponse uses toUploadUrl (path and url become /api/uploads/<folder>/<file>); the subfolder-by-file-type logic was removed.
- deleteUpload resolves the path with getLocalUploadPath (both forms, traversal-safe) and checks references against both forms of the path in every table (Prisma { in: [...] }; settings use an OR of contains).
- Unused imports (path, UPLOADS_DIRECTORY) removed.

src/controllers/settingController.js

- uploadSettingFile stores toUploadUrl(file), i.e. /api/uploads/csr-policy/... or /api/uploads/printers/..., instead of the hard-coded /uploads/documents/....
- removeLocalFile uses getLocalUploadPath, so old files at either path form are cleaned up.
- Unused imports removed.

src/controllers/noticeController.js

- Local helper getLocalDocumentPath removed in favour of getLocalUploadPath; getUploadedDocumentUrl uses toUploadUrl.
- New helper placeNoticeDocument(url, type, category) moves the document to its folder and returns { url, undo }.
- createNotice and updateNotice place the document, write the row inside try/catch, and undo the move if the write fails (section 4.4).
- updateNotice deletes the previous document only when a different document was sent.
- Imports rename; path import removed.

src/controllers/sectionController.js

- Local helpers getLocalFilePath and toUploadUrl removed; uses the shared getLocalUploadPath and toUploadUrl.
- New helper moveReqFilesToModuleFolder; called in createSection and updateSection before paths are built (section 4.5).
- path import removed.

src/validators/uploadValidators.js

- deleteUploadValidator accepts paths starting with /api/uploads/ or /uploads/; message now says /api/uploads/.

src/validators/common.js

- Comment example updated to /api/uploads/leaders/x.webp.

src/routes/uploadRoutes.js

- New folderFromQuery = setUploadFolder(req => req.query.folder), added before multer on the image, document and video routes.
- Swagger: folder query parameter on the three POST routes, description text updated, DELETE example path updated.

src/routes/settingRoutes.js

- POST /admin/settings/:key/file runs setUploadFolder(req => SETTING_UPLOAD_FOLDERS[req.params.key]) before multer.

`src/routes/noticeRoutes.js`

- POST / and PUT /:id run `setUpUploadFolder("notices")` before `uploadPdf.single("document")`.

`src/routes/sectionRoutes.js`

- POST /admin/sections and PATCH /admin/sections/:id run `setUpUploadFolder()` (others) before `uploadSectionFiles`.

`src/routes/dashboardRoutes.js`

- Swagger YAML fix: summary of PUT /api/admin/distribution/{year} is quoted.

`src/utils/landingPage.js`

- Swagger button links to /api/docs.

`README.md`

- /api-docs changed to /api/docs (2 places).

`package.json`

- New script: "organize-uploads": "node scripts/organize-uploads.js".

`scripts/organize-uploads.js`

- New: migration script (section 5).

Not changed

- Prisma schema and migrations, seed data.
- Paths, request bodies and response shapes of business endpoints. Only the path values inside responses change.
- Authentication, rate limiting, CORS configuration, validators other than those listed.
- Environment variables. No new variables.

7. API contract changes for clients

- **Base URL:** clients use the bare host (for example `https://bstbpc.bihar.gov.in`) and call paths starting with `/api/`. The frontend's `VITE_API_URL` should not end in `/api`.
- **Swagger:** `<host>/api/docs`.
- **Uploading:** `POST /api/uploads/{image|document|video}?folder=<name>` with multipart field `file`. Response: `{ success, data: { path, url, filename, mimeType, size } }`; `url` is `/api/uploads/<folder>/<file>`.
- **Showing files:** link to `<host>/api/uploads/...` For older `/uploads/...` values, add `/api` in front (the frontend's `fileUrl()` does this).
- **Deleting files:** `DELETE /api/uploads` with `{ "path": "/api/uploads/..." }`; the old `/uploads/...` form is also accepted.
- **Errors:** unknown folder gives `400 { success: false, message: "folder must be one of: ..." }`; unknown `/api` route gives `404 JSON`.

Frontend changes that go with this (for reference)

- `services/api.js`: new `isUploadedFile()`; `fileUrl()` handles `/api/uploads/` and legacy `/uploads/`.
- `services/uploadService.js`: `uploadFile(file, kind, folder)` sends `?folder=`; folder constants, `moduleUploadFolder()` and `noticeUploadFolder()`.
- Each admin page sends its folder: Notices (by type/category), Books (covers, chapters), Leaders, Tools & Resources / Latest Initiatives, CSR, Website editor (by module) and the MD photo.
- Video gallery and website editor use `isUploadedFile()` to detect uploaded videos.
- The frontend and backend must be deployed together: the old frontend does not send `?folder=`, so its uploads would all go to others/, and it builds `/uploads/...` links that do not work on the hosted server.

8. Testing performed

All tests ran on 24 Sep 2026 against a local server (port 3999) using the local development MySQL database, unless marked production.

Part A

Check	Result
<code>node --check</code> on <code>server.js</code> , all controllers, <code>uploads.js</code> , <code>validators</code> , <code>swagger.js</code> , the script	All pass
<code>GET /api, /api/test, /api/docs/</code>	200
<code>GET /api/books?limit=1, /api/sections?module=gl-photo, /api/getDirectory/leader, /api/notices</code>	200
<code>GET /api/uploads/images/<file>, /api/uploads/image/<file></code>	200
<code>GET /uploads/images/<file>, GET /</code>	404 (intended)
<code>GET /api/nope</code>	404 JSON
Swagger spec	31 paths, all start with <code>/api/</code> ; dashboard endpoints present; <code>DELETE /api/uploads</code> has no folder parameter
Path helpers: new form, legacy form, <code>/api/uploads/../../etc/passwd</code> , external URL, <code>/images/x</code>	Resolved, resolved, null, null, null
Production: <code>/uploads/images/<leader photo></code>	404, confirms the cause of the broken images
Production: <code>/api/uploads/images/<leader photo></code>	200 image/png

Part B (admin JWT signed locally for an existing admin)

Step	Result
1. POST /api/uploads/document?folder=tenders	Saved as /api/uploads/tenders/<file>.pdf
2. POST /api/uploads/document?folder=../etc	400: folder must be one of ...
3. POST /api/uploads/document (no folder)	Saved as /api/uploads/others/<file>.pdf
4. Upload with folder=notices, then create a notice with category Circular using that path	Stored as /api/uploads/circulars/<file>.pdf; GET 200
5. Update the notice to type Tender (no new file)	Moved to /api/uploads/tenders/<file>.pdf; GET 200; old circulars path 404
6. Delete the notice	200; document file deleted (GET 404)
Clean-up	Test notice deleted by step 6; test files from steps 1 and 3 removed
organize-uploads dry run	1 file to move, 1 row to update, 11 unreferenced files (section 5)

Not tested: the migration script with --apply; the multipart notice/section upload paths with real files (same helpers as the tested paths); the settings file upload route end-to-end.

9. Deployment checklist

- 1 Back up the MySQL database and the backend's uploads/ directory.
- 2 Deploy the backend code. No new dependencies, no Prisma migration, no new environment variables.
- 3 Restart the backend. On start it creates the 23 folders under uploads/.
- 4 Check <https://bstbpc.bihar.gov.in/api/test> (200 JSON), /api (status page) and /api/docs/ (Swagger).
- 5 Deploy the frontend at the same time (see section 7).
- 6 On the server, from the backend folder: `npm run organize-uploads` and review the output.
- 7 If the plan looks right: `npm run organize-uploads -- --apply`.
- 8 Open the website and admin panel: leader photos, book covers, notices/tenders PDFs, gallery, CSR PDF. Upload one test file per admin page and check which folder it lands in.
- 9 Optional: once nothing references them, remove the singular aliases `/api/uploads/image|document|video` from `server.js` and review the unreferenced files the script listed.

Reverse proxy: only `/api/*` needs to be forwarded to the backend. Any separate rule for `/uploads/*` is no longer needed.

10. Known issues and items not done

- Production is still on the old backend and frontend. Images uploaded there are broken until both are redeployed.
- The notices and directory endpoints work but are not described in Swagger (this was already the case before these changes).
- Production data: the leader list has one test entry ("Aman", designation "God"); the four real leaders are in the local database but not in production.
- Files moved when a notice's type or category changes are moved even if another row somehow references the same file. This does not happen through the admin panel, which uploads a new file per notice.
- Files that no row references are reported, not moved or deleted.
- If the frontend sends no folder, files go to `others/`. Routes without an upload screen yet (officers, employees, admins) are covered only by the migration script.
- In the frontend, lint still reports 7 errors in the admin pages that were edited: unused motion imports and a regex escape in `BooksPage.jsx`. None are on changed lines, and the build succeeds.